

SALTDRAW · DESIGN DECISIONS REGISTER

SaltDrag design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/saltdrag		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened at TRL 3; every decision made under Amish's 2026-10-03 pre-approval

SaltDrag design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All decisions were made under Amish's 2026-10-03 pre-approval.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The rope certificate shows a minimum breaking strength of at least 25 kN for the 12 mm polyester double braid, with eye splices	The 7.7 working-load factor (R10) rests on it	SDG-CAL-001 [C8]
2	The bought gear and pinion are module 4, 20 degree pressure angle, C45 or better, and the gear blank is webbed (about 10 kg)	Tooth strength, mesh at 196 mm centres and the part mass	SDG-CAL-001 [C3]; SDG-DDR-002
3	The flange bearings fit 40 mm and 25 mm shafts on bolt squares of 102 mm and 70 mm, with stainless inserts	The cheek holes are drilled to these squares	SDG-DWG-101
4	The chain, turnbuckles and shackles carry working load limits of at least 7.8 kN, 6 kN and 10 kN	The 2.4 margin on the chain line	SDG-CAL-001 [C9b]
5	The FRP tube's maker gives a flexural strength of 200 MPa or more	The beam margin of 7.1	SDG-CAL-001 [C13b]
6	Each helical anchor holds at least 2.25 kN (1.5 times its share of the rated pull) in the proof pull on site, at the trial pan	The soil strength is assumed at 50 kPa; the anchors are a safety stop	SDG-CAL-001 [C10]; SDG-DDR-001, D11
7	The drag of the rake on the partner's crystal bed, measured with a spring scale or CalRig on a short trial rake	The design drag and rope pull rest on an assumed 300 N per metre; if higher, raise the skids or shorten the rake	SDG-CAL-001, Section A
8	The bund crest is at least 0.35 m wide and the ground behind the bund is firm and dry on both capstan sides of the trial pan	The roller saddle footprint and capstan position assume it	SDG-PRB-001; SDG-DDR-001, D2

■ Value engineering

Value-engineering target: USD 4,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 2,803 (USD 1,197 under the target). Main cost drivers and savings worth trying:

- The largest lines are fabrication labour and galvanising (USD 450), the four side frames (USD 220), the two drums (USD 220), the two bought gears (USD 190), the rake blade (USD 190) and the two pull ropes (USD 190).

- The four flange bearings with stainless inserts (USD 264 for eight) cost about twice plain cast-iron bearings; they stay because the capstans stand in salt dust and are rinsed daily.
- Savings worth trying: a single gear pair cut in a local workshop in place of bought gears; a laser-cut cheek nested with the ratchet wheel and pawl to cut plate waste; sharing one set between several families' pans, which spreads the cost per family.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	Geared drum winch (still called a hand capstan) rather than a warping capstan	Amish, pre-approval: "I pre-approve the batch runs along with any recommendations you come up with."	SDG-DDR-001, D1
2026-10-03	Capstan on dry ground 1.5 m behind the bund crest, with a bund roller saddle on the crest	Amish, same pre-approval	SDG-DDR-001, D2
2026-10-03	Single spur stage 6 to 1, module 4; two 330 mm cranks 180 degrees apart	Amish, same pre-approval	SDG-DDR-001, D3
2026-10-03	12 mm polyester double braid rope, 45 m per side	Amish, same pre-approval	SDG-DDR-001, D4
2026-10-03	One-piece 3.0 m FRP rake with a 300 mm UHMW-PE blade; three adjustable skids, blade 10 mm above the floor	Amish, same pre-approval	SDG-DDR-001, D5 and D6
2026-10-03	Two helical ground anchors per capstan, chained at rope height	Amish, same pre-approval	SDG-DDR-001, D7
2026-10-03	Galvanised steel on dry ground; FRP, UHMW-PE and 316 stainless in the pan; stainless-insert bearings; daily rinse	Amish, same pre-approval	SDG-DDR-001, D8
2026-10-03	Safety: pawl always engaged while winding; pay out only on the brake with the cranks removed; rope dampers and a 2 m exclusion zone; anchors proof-loaded on site to 1.5 times their share	Amish, same pre-approval (conservative options)	SDG-DDR-001, D9 to D11
2026-10-03	Design pan 30 m between crests; crew of four	Amish, same pre-approval	SDG-DDR-001, D12 and D13
2026-10-03	First co-design partner to approach: SEWA; first trial region: Little Rann of Kutch (candidates, not agreed)	Amish, same pre-approval	SDG-DDR-001, D14 and D15
2026-10-03	Seek written clarification from the forest department, through the partner, that a hand capstan is not a machine	Amish, same pre-approval	SDG-DDR-001, D16
2026-10-03	SaltDrag capstan offered as the first candidate for the hand capstan block shared with SiltHaul	Amish, same pre-approval	SDG-DDR-001, D17
2026-10-03	Design for construction: bolted side frames and cross tubes, welded drum with brake rim, flange bearings, bought gears, pawl on a welded boss, differential band brake, removable cranks, gear guard, anchor ears and chains, one-piece rake with crush-sleeved bolts, slotted skid hangers, end caps with eye plates	Amish, same pre-approval	SDG-DDR-002, changes 1 to 14
2026-10-03		Amish, same pre-approval	SDG-DDR-002,

Date	Decision	Decided by	Record
	Ballast bags on the rake; windrow board removed (the windrow forms about 3.3 m from the bund toe); rope dampers added		changes 15 to 17
2026-10-03	Estimated cost reported against the unchanged value-engineering target of USD 4,000	Amish: "I also accept any cost overruns or variations from the assumed scope cost."	This register